

CyberInsight AI: Intelligent Cyber Threat Analysis Platform

Domain: Artificial Intelligence and Cybersecurity

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Rapid Growth of Cybercrime

Explosive Adoption

Widespread smartphone and high-speed internet adoption have triggered an exponential surge in mobile and web cyber threats.

Unverified Channels

Users receive malicious APK files, executables, phishing links, and scam QR codes via WhatsApp, Telegram, email, and social media.

Total Compromise

Unknowing installation leads to device compromise, sensitive data theft, identity fraud, and direct financial losses.

The CyberInsight AI Solution

Bridges the security gap by converting cryptic threat indicators into AI-assisted explanations and actionable recommendations in simple language.



Mobile Threats

Malicious APK payloads targeting Android devices detected before installation.



Phishing Links

Deceptive web portals spoofing legitimate services flagged and explained.



Scam QR Codes

Malign QR redirects intercepted before users are exposed to harmful payloads.



AI Defense

Human-readable advice empowering non-technical users to act safely and confidently.

Core Problem

Everyday users cannot identify whether a downloaded file or application is safe prior to installation or execution. Existing security solutions output complex forensic reports requiring professional cybersecurity knowledge.

⚠ There is no beginner-friendly platform that translates security risk into clear, conversational language.

Major Attack Vectors Analyzed

Fake Banking APK

Steals credentials &
OTPs

Fake e-Challan APK

Trojanized traffic apps

Malicious EXE

Executable malware
payloads

Phishing Websites

Spoofed credential
portals

Scam QR Codes

Malicious payment
links

The Platform Gap

No existing solution combines multi-format inspection with plain-language AI explanations for non-technical users.

Existing Solutions

VirusTotal

Aggregates 70+ antivirus engines, outputs raw detection ratios and complex technical reports.

Google Play Protect

On-device Android scanner for installed apps — minimal explanation provided.

MobSF

Automated APK security framework designed for security researchers — complex interface.

Key Systemic Limitations

1

Cryptic Reports

Output is unreadable without cybersecurity expertise.

2

Expert-Only UX

Interfaces designed for professionals, not everyday users.

3

Tool Fragmentation

No unified platform covering APKs, EXEs, and URLs together.

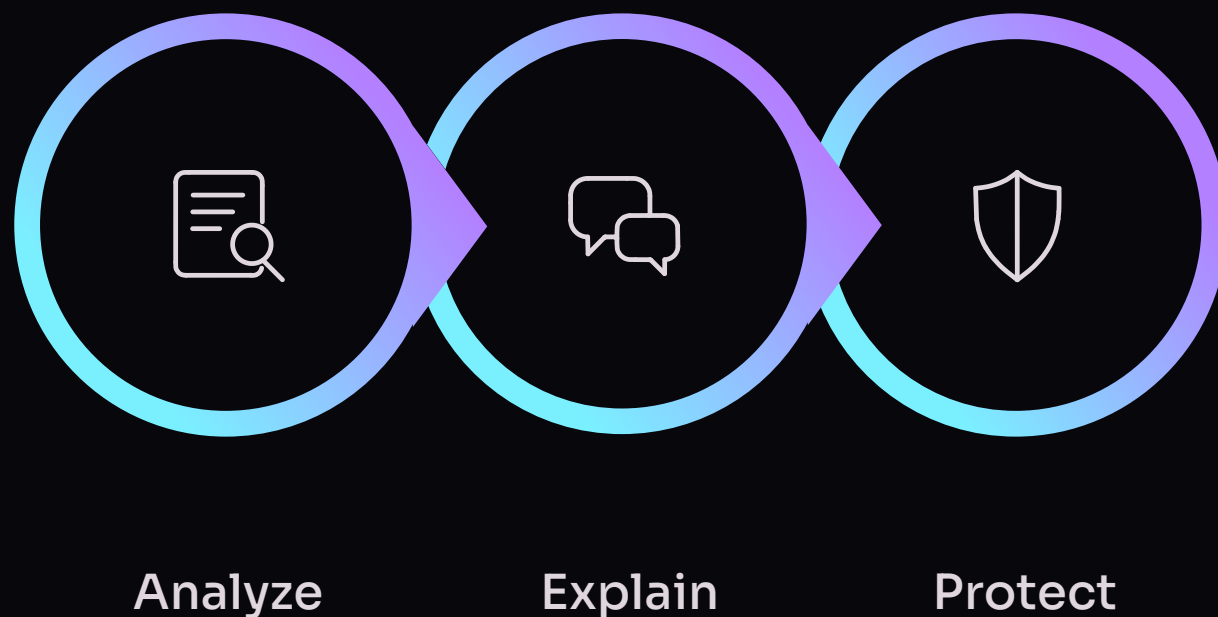
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No AI Explanation

Raw detection data is never translated into actionable guidance.

Pre-Execution Static Analysis

CyberInsight AI executes multi-layered static analysis on suspicious files **prior to installation or execution**, ensuring zero sandbox infection risk.



Multi-Format Inspection

- APK Files — Android mobile packages
- EXE Files — Windows executables
- Web URLs — Phishing & scam links

AI Intelligence Triad Output

- **Risk Score** — Low, Medium, or High
- **Threat Summary** — Plain-language narrative
- **Advice** — Install, Block, or Delete

Project Objectives

01

Static APK Analysis

Parses AndroidManifest.xml without execution

02

Static EXE Analysis

Inspects PE headers, metadata, & binaries

03

SHA-256 Hashing

Generates hash fingerprints for verification

04

Permission & Certificate Audit

Detects dangerous privileges (SMS, Camera) and validates developer signatures

05

URL Analysis

Scans HTTPS, typosquatting, & phishing indicators

06

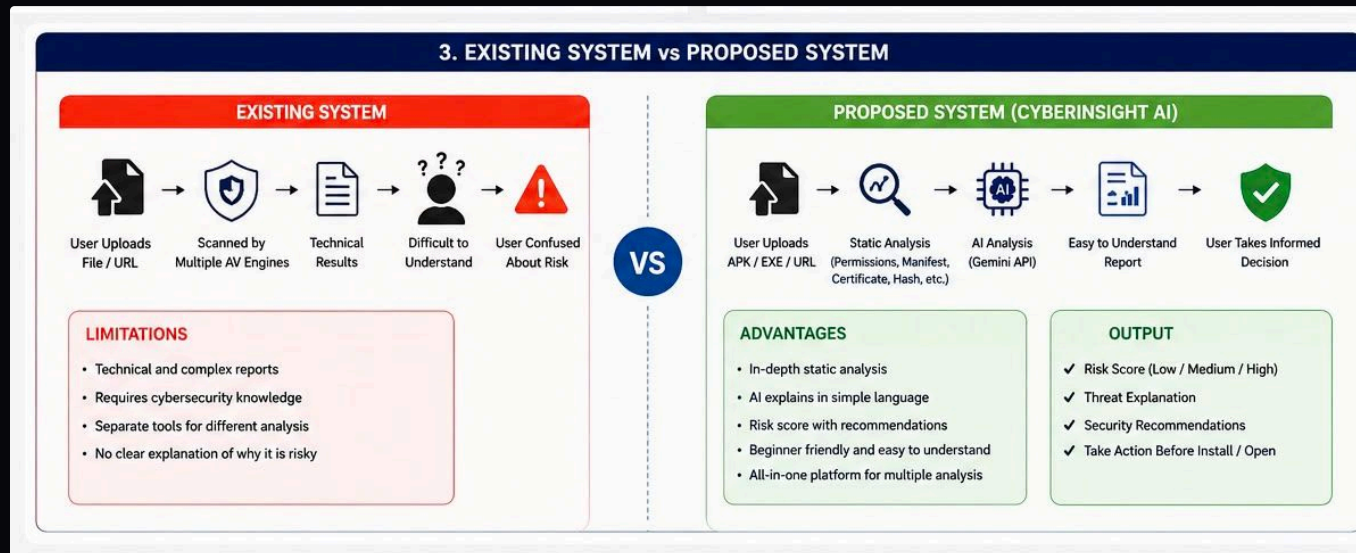
AI Explanation & Awareness

Translates raw logs into plain language, empowering non-technical users



Primary Goal: Empower non-technical users to make informed, safe decisions before installation.

System Workflow Architecture



Module 1: APK Analysis

- **AndroidManifest:** Parses components and intent filters
- **Permissions:** Flags dangerous privileges (SMS, Camera, Contacts)
- **Certificate:** Checks developer signature validity & trust
- **SHA-256 Hash:** Fingerprints APK for malware matching

Module 3: URL Analysis

- **HTTPS Verification:** Audits SSL/TLS encryption state
- **Suspicious Keywords:** Detects phishing triggers (login, verify)
- **Domain Information:** Evaluates WHOIS, domain age & typosquatting

Module 4: AI Report

- **Risk Score:** Synthesizes data into Low, Medium, or High Risk
- **Threat Summary:** Plain-language narrative of structural findings
- **Recommendations:** Direct user advice (Install, Block, or Delete)

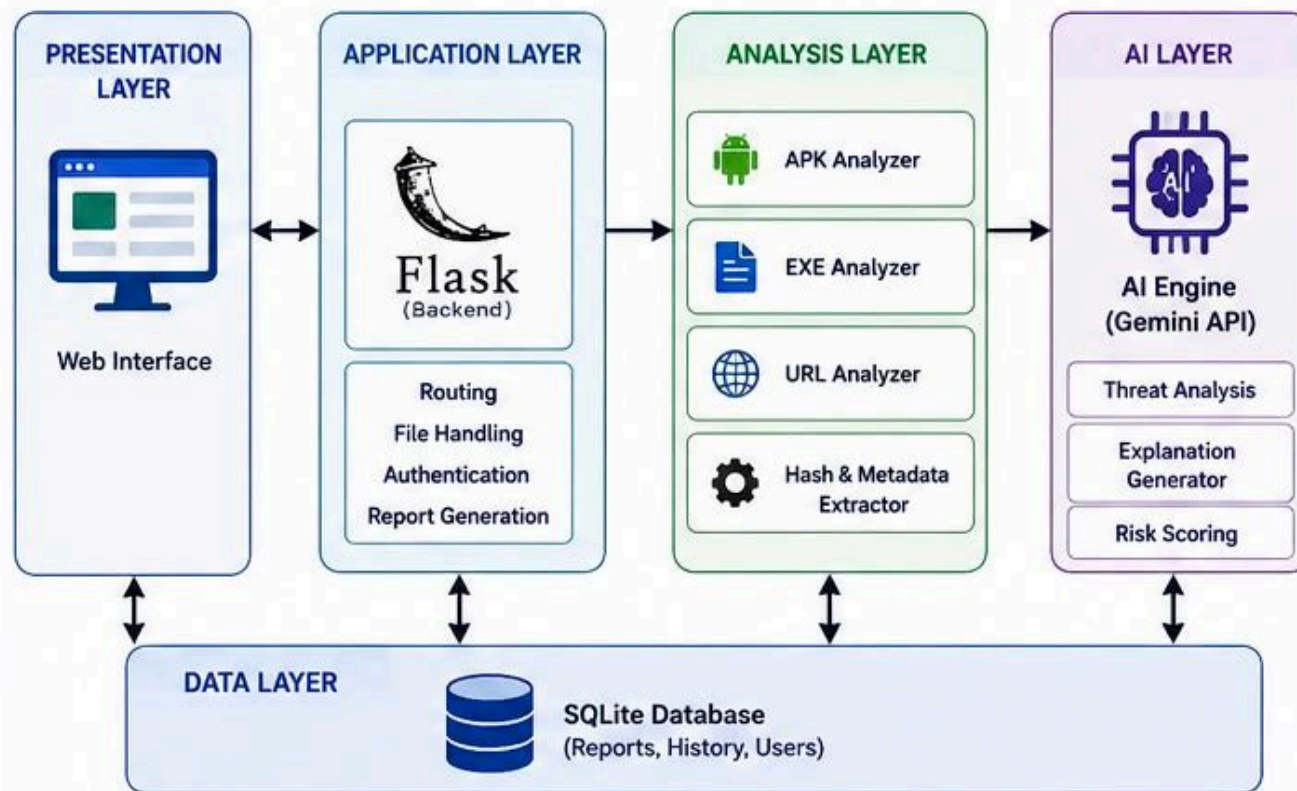
Module 2: EXE Analysis

- **Metadata:** Inspects PE headers, timestamps, & compiler info
- **Digital Signature:** Verifies publisher identity and integrity
- **Suspicious Strings:** Scans for embedded IPs, registry keys & APIs

Static analysis is performed without executing the uploaded file, making the process safer and faster.

Software Stack Specifications

2. SYSTEM ARCHITECTURE DIAGRAM



Technology Stack

Frontend

HTML5, CSS3, JavaScript (ES6)

Backend

Python 3.11, Flask Framework

Database

SQLite — Reports, User Logs, History

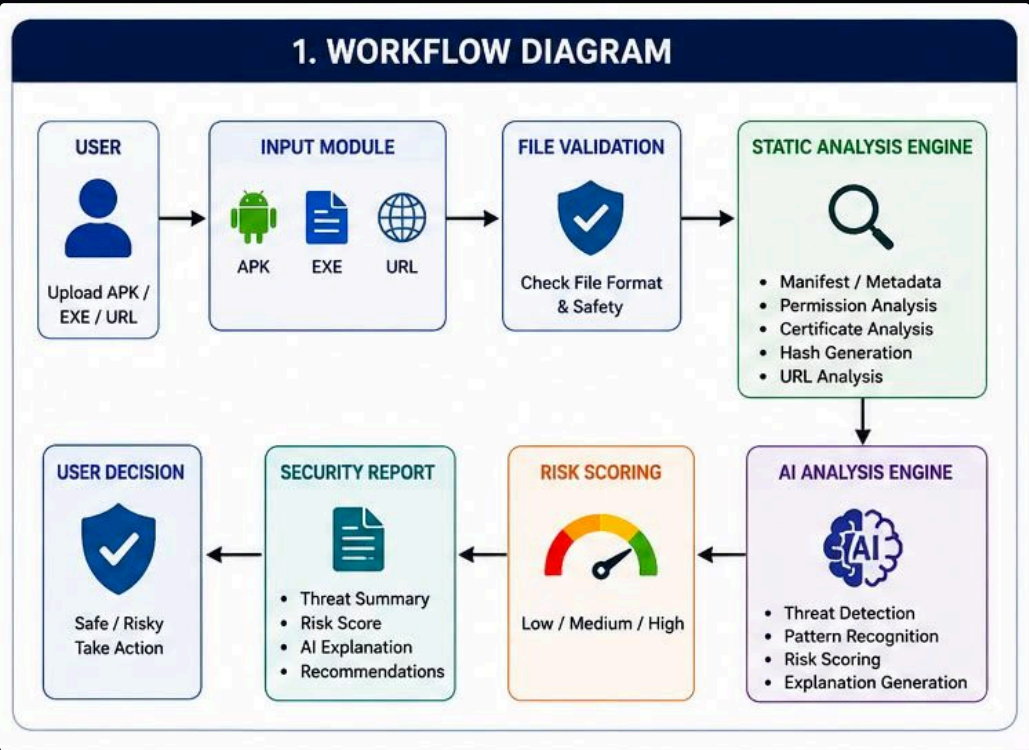
AI Engine

Gemini API Integration

Libraries

hashlib, androguard, requests, os

Comparison: VirusTotal vs CyberInsight AI



❏ **Strategic Positioning:** CyberInsight AI complements existing tools instead of replacing them — helping users understand *why* a file may be risky through AI-assisted explanations.

Feature	VirusTotal	CyberInsight AI
Analysis Engine	Multiple Antivirus Engines	Static Analysis (APK & EXE)
Output Type	Technical Detection Logs	AI Plain-Language Explanation
Core Focus	Detection Focused	Detection + Explanation
User Audience	Requires Technical Skill	Beginner Friendly
Actionability	Raw Malware Counts	Risk Score & Recommendations

Key Project Innovations



AI-Assisted Threat Explanation

Automated conversion of binary signatures and forensic indicators into clear human narrative.



Beginner-Friendly Interface

Clean, intuitive UX tailored specifically for general computer users — no expertise required.



Unified Multi-Module Platform

Single scanner handling APKs, EXEs, and Web URLs — eliminating tool fragmentation.



User Cyber Awareness

Educates users on permission abuse and threat risks, building long-term security literacy.